

INSTITUTE OF SPACE TECHNOLOGY
KICSIT, Kahuta Campus



# APP DEVELOPMENT

## ASSIGNMENT NO 1 (LAB)

## SUBMITTED TO:

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## Table of Contents


- 1. Introduction
- 2. Objectives
- 3.Tools and Technologies
- 4. Reference Application analysis
- 5. Development Environment Setup
- 6. TensorFlow Lite Model Integration
- 7. Testing and Result
- 8. Screenshots and Evidence
- 9. Challenges and solution
- 10. Conclusion
- 11. References

# 1 Introduction

Artificial Intelligence (AI) allows computers to perform tasks such as recognizing images and predicting results. Machine Learning (ML) is a part of AI that learns patterns from data.

TensorFlow Lite (TFLite) allows machine learning models to run on mobile devices. It helps Android applications perform predictions directly on a device.

In this assignment, I studied the instructor-provided reference application and worked with an Android application named NumberGuessing App TFLite. The application allows users to draw a handwritten digit and displays the predicted digit and its confidence score.

# 2 Objectives

The main objectives of this assignment are:

- To understand how AI can be integrated into Android applications.
- To study the provided reference application and its source code.
- To understand the use of TensorFlow Lite for on-device prediction.
- To develop and run an Android application that recognizes handwritten digits.
- To test the application with different inputs.
- To document the implementation with screenshots and testing results.

# 3 Tools and Technologies

The following tools and technologies are used in this project:

- **Android Studio:** Used to develop and run the Android application.
- **Java:** Used for the application's source code.
- **TensorFlow Lite:** Used to run the machine learning model.
- **Gradle:** Used to manage project configuration and dependencies.
- **Android Emulator or Physical Device:** Used to run and test the application.
- **GitHub:** Used to store and share the source code.

# 4 Reference Application Analysis

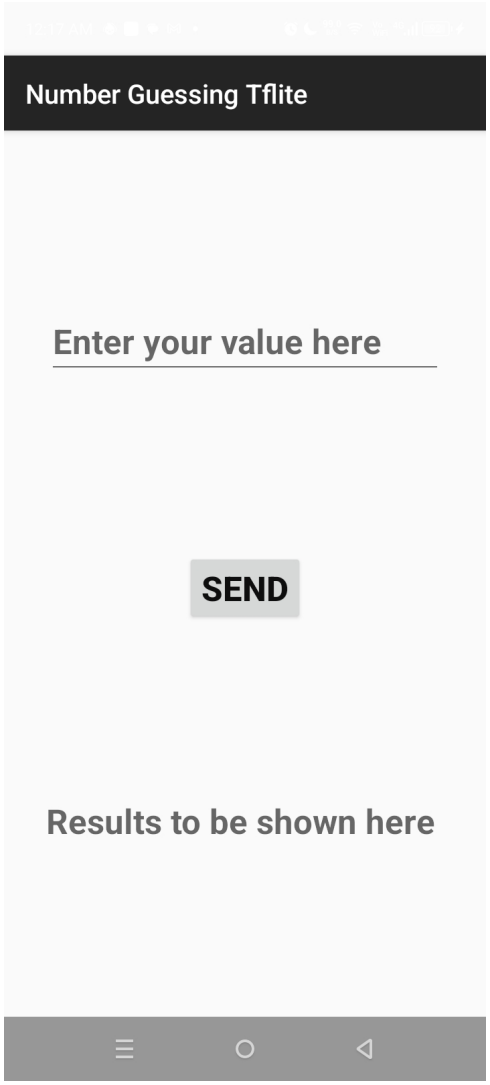
The instructor-provided reference application was used to understand the structure and working of an AI-based Android application.

The reference project was examined to identify its main source files, user interface, model files, and dependencies.

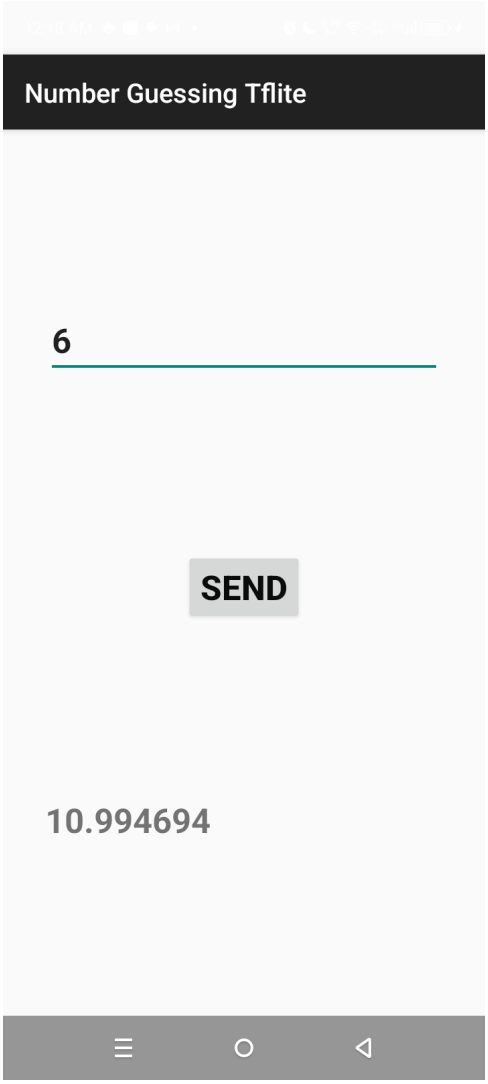
The project structure included Java source files, Android resources, the Android manifest, Gradle configuration files, and a TensorFlow Lite model.

The reference application was tested to observe its behavior and understand how it processes user input and displays predictions.

## 4.1 Reference Application Main Screen



### 4.2 Reference Application — Input and Prediction Result




## 5 Development Environment Setup

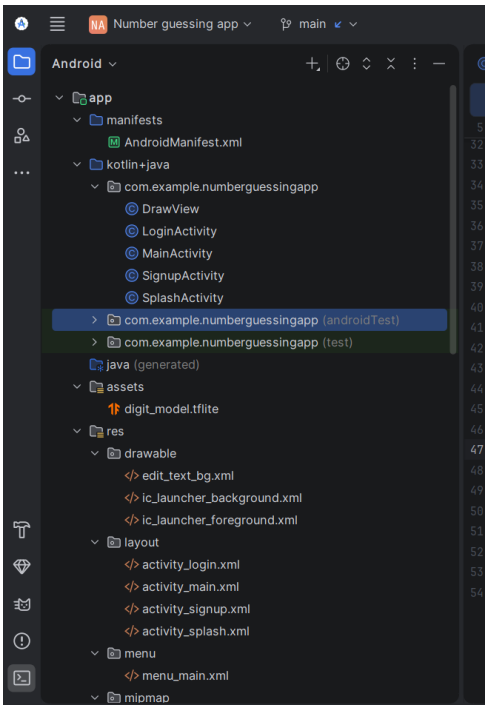
Android Studio was used as the development environment.

The project was opened in Android Studio, and the Gradle configuration was checked. The application was then built and run on a compatible Android device or emulator.

The following steps were followed:

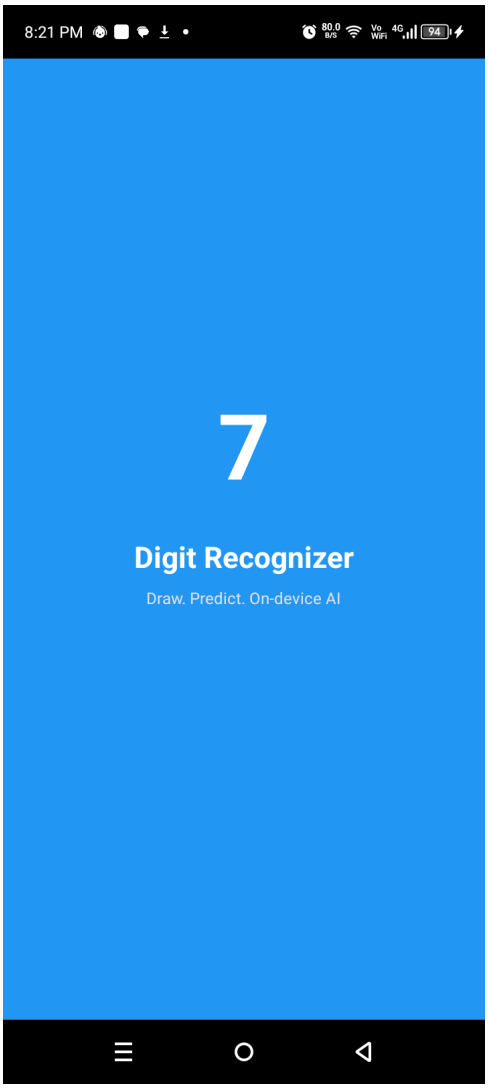
1. Open Android Studio.
2. Open the provided or developed Android project.
3. Wait for Gradle Sync to complete.
4. Check the required dependencies and model files.
5. Select an Android device or emulator.
6. Click the Run button.
7. Confirm that the application starts successfully.

## 5.1 Android Studio Project Structure

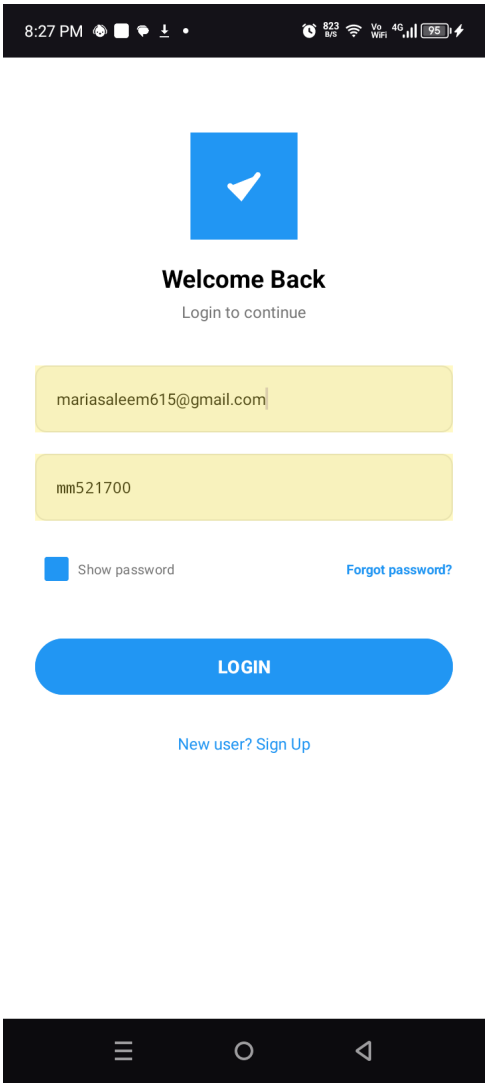


## 5.2 Application Running on Device or Emulator

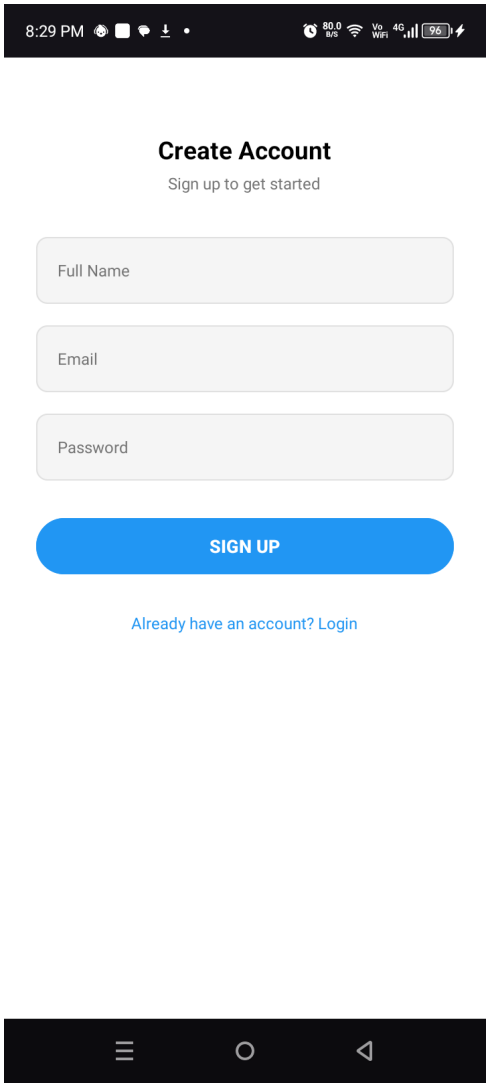
### 5.2.1 Splash Screen



### 5.2.2 Login Screen



### 5.2.3 Signup Screen




# 6 Implementation of NumberGuessing App TFLite

The application was developed using Java in Android Studio.

The user can draw a handwritten digit on the drawing area. The application processes the drawing and passes the input to the machine learning model.

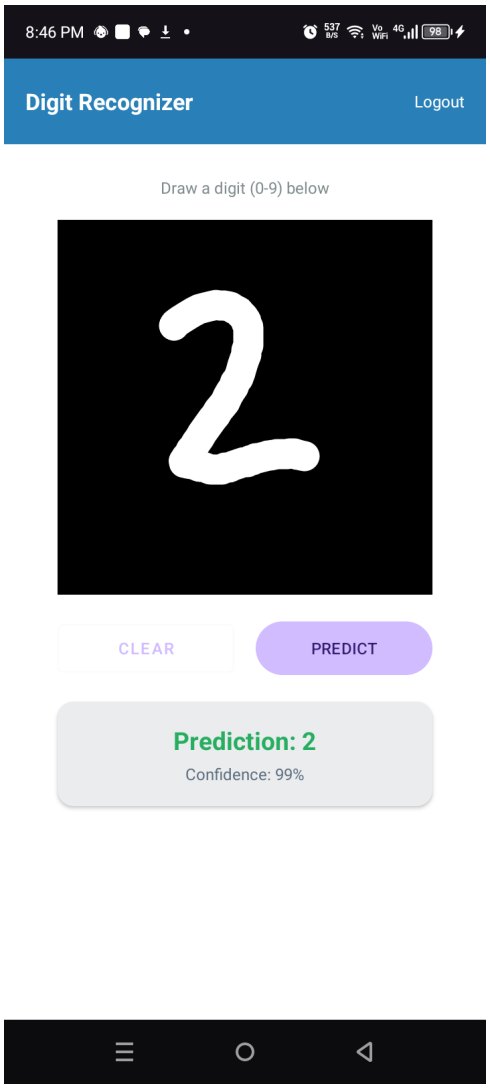
The model predicts the digit, and the application displays the result and a confidence score.

The main project files include:

- `MainActivity.java`: Main application activity.
- `DrawView.java`: Drawing interface for user input.
- `LoginActivity.java`: Login screen.
- `SignupActivity.java`: Signup screen.
- `SplashActivity.java`: Splash screen.

- `digit_model.tflite`: TensorFlow Lite model.
- `AndroidManifest.xml`: Application and activity configuration.

## 6.1 Drawing Screen of My Application




# 7 TensorFlow Lite Model Integration

TensorFlow Lite is used to run a trained machine learning model on an Android device.

The project contains a model file named `digit_model.tflite` in the assets folder.

The application uses the model to process the drawn digit and generate a prediction. The output is interpreted by the application and displayed to the user.

The general process is:

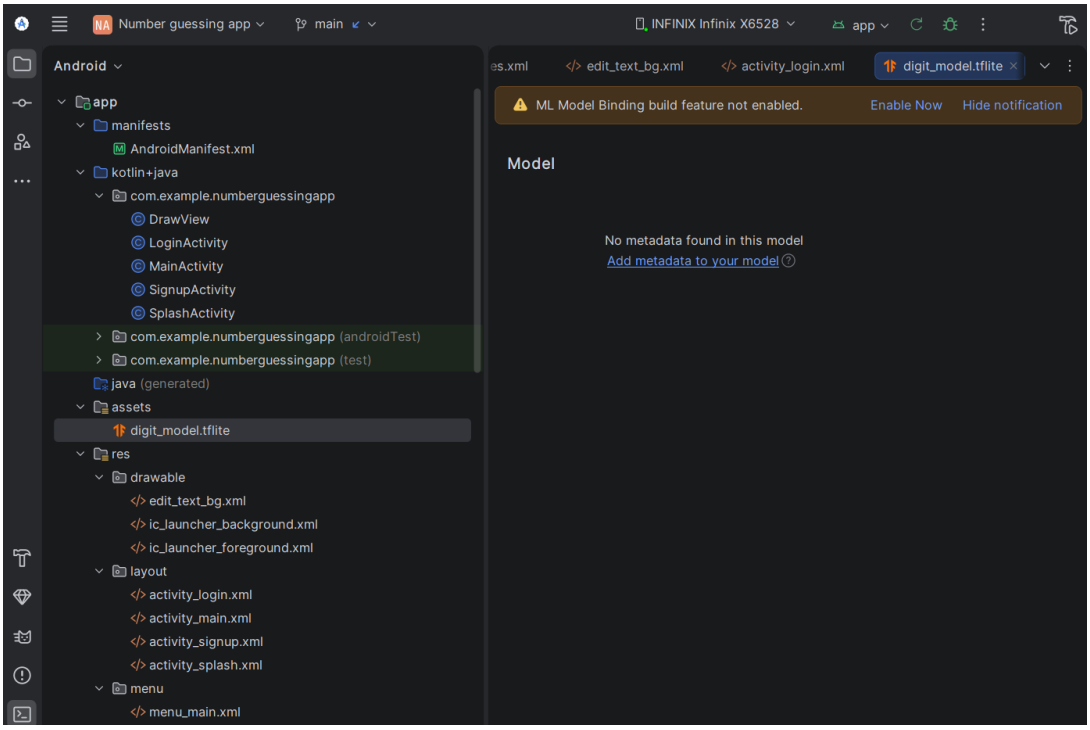
1. The user draws a digit.
2. The application prepares the drawing as model input.
3. The TensorFlow Lite model processes the input.

4. The application interprets the model output.

5. The predicted digit and confidence score are displayed.

The exact input dimensions, output format, and prediction calculation depend on the integrated model and source code.

## 7.1 digit_model.tflite Inside the Assets

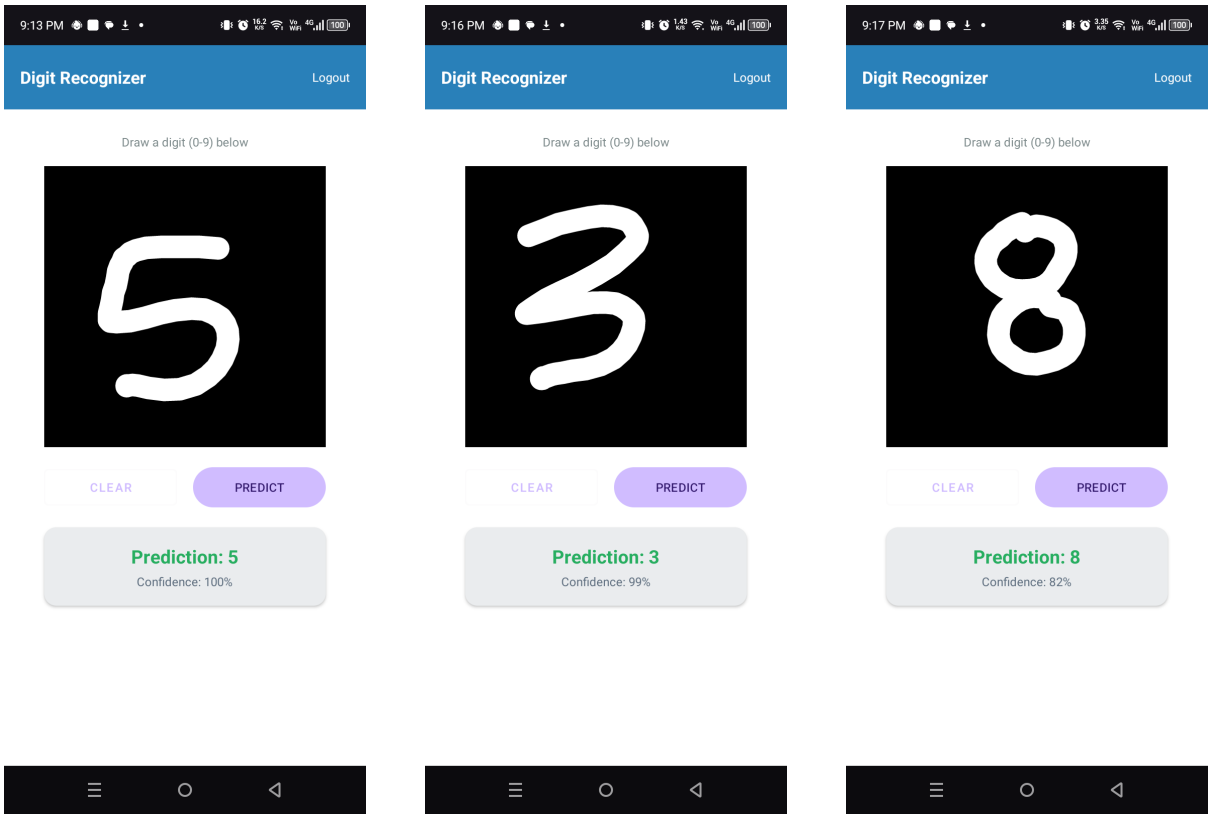



# 8 Testing and Results

Testing was performed to check whether the application accepts handwritten digits and displays predictions.

Both the instructor-provided reference application and the developed application should be tested separately.

## 8.1 Test Results Table




| Test Case | Input | Expected Result | Actual Result |
|---|---|---|---|
| 1 | Draw digit 5 | Display the correct digit | 100% |
| 2 | Draw digit 3 | Display the correct digit | 99% |
| 3 | Draw digit 8 | Display the correct digit | 82% |

# 9 GitHub Repository and Source Code Management

GitHub is an online platform used to store, manage, and share source code. It helps developers keep their project files in one place and track changes made to the code.

For the NumberGuessing App TFLite project, GitHub was used to upload and maintain the Android application source code. The repository contains the project files, Java source code, Android configuration files, and the TensorFlow Lite model file (`digit_model.tflite`).

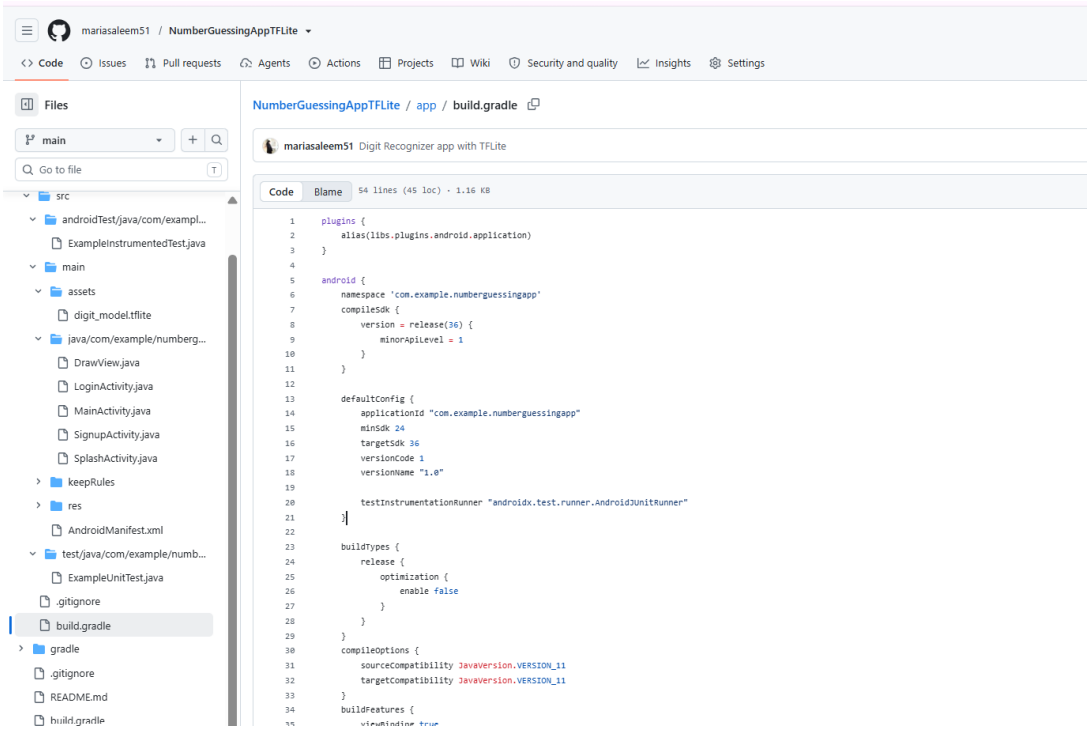
GitHub makes the project easier to access, review, and share with teachers or other developers. It also provides a backup of the source code and helps developers collaborate on software projects.

## 9.1 Benefits of GitHub

1. **Code Storage:** Stores project files in an online repository.
2. **Version Control:** Helps track changes made to source code.

3. **Backup:** Keeps a remote copy of project files.
4. **Sharing:** Allows teachers and developers to view the project.
5. **Collaboration:** Enables multiple developers to work on the same project.

## 9.2 GitHub Repository and Uploaded Source Code




# 10 GitHub Repository Cloning Procedure

## 10.1 Introduction

GitHub cloning is the process of copying a repository from GitHub to a local computer. It allows developers to access the source code and work on the project using Android Studio.

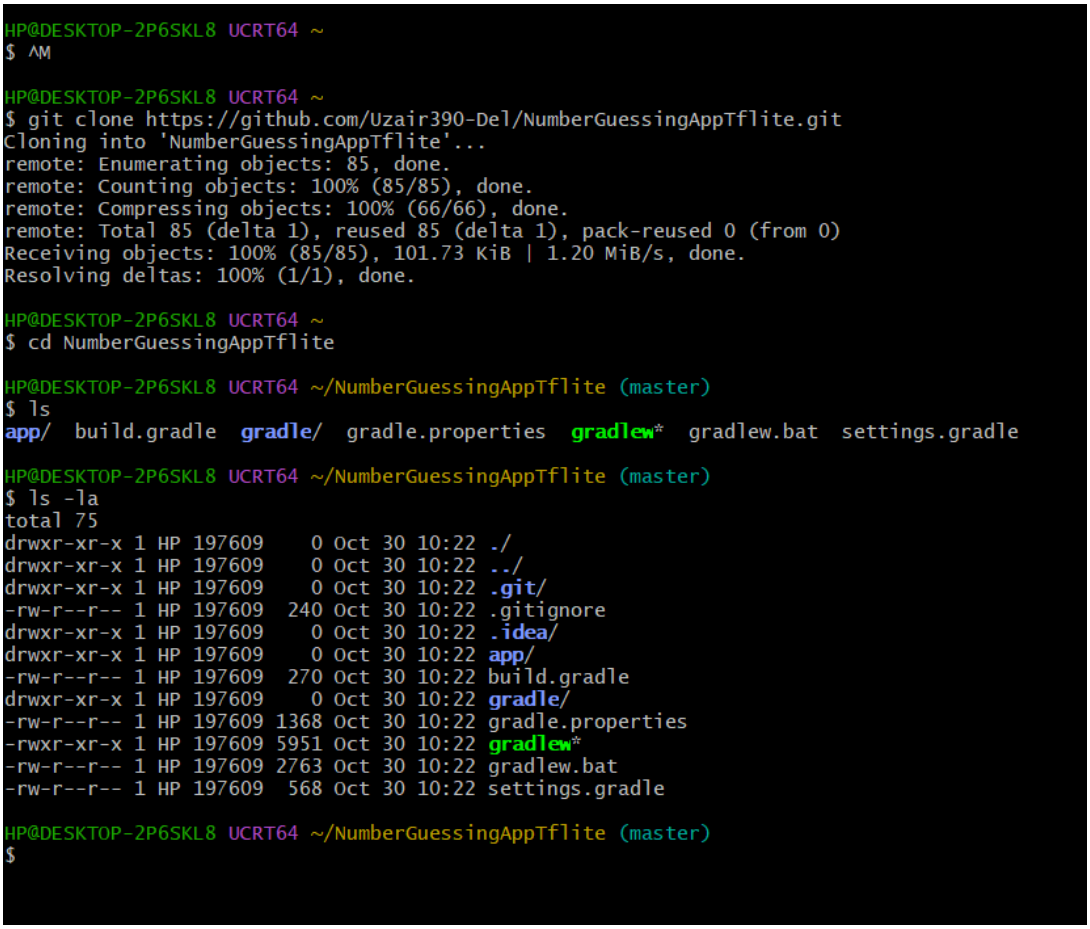
## 10.2 Steps for Cloning the Repository

1. Open the GitHub repository in a web browser.
2. Click the green Code button.
3. Copy the HTTPS repository link.
4. Open Command Prompt or Terminal on the computer.
5. Type the `git clone` command followed by the copied repository link.
6. Press Enter to download the repository.
7. Open Android Studio and select the cloned project folder.

8. Wait for Gradle Sync to complete.
9. Check the project files and folder structure.

## 10.3 Git Command

`git clone <repository-link>`




# 11 Repository Structure and Explanation

The repository contains different files and folders. Each has its own purpose.

1. `app/` – Contains the main Android application files.
2. `build.gradle` – Contains project build settings and dependencies.
3. `gradle/` – Contains Gradle wrapper files.
4. `gradle.properties` – Contains Gradle configuration settings.
5. `gradlew` – Runs Gradle commands on Linux and Mac.
6. `gradlew.bat` – Runs Gradle commands on Windows.
7. `settings.gradle` – Defines the project name and included modules.

8. `.gitignore` – Tells Git which files should not be tracked.
9. `.idea/` – Contains Android Studio settings.
10. `.git/` – Stores Git history and repository information.

### 11.1 Commands Used

1. `git clone` – Downloads the GitHub repository to the computer.
2. `cd NumberGuessingAppTflite` – Opens the cloned project folder.
3. `ls` – Shows the files and folders in the repository.
4. `ls -la` – Shows all files and folders, including hidden files.

## 12 Conclusion

This assignment provided practical experience in Android application development and TensorFlow Lite integration.

The NumberGuessing App TFLite project demonstrates how a user can draw a handwritten digit and receive a model prediction on an Android device.

The project also helped

## 13 References


1. TensorFlow Lite documentation: https://www.tensorflow.org/lite
2. Android Developers documentation: https://developer.android.com/
3. Instructor-provided reference repository: https://github.com/mariasaleem51/NumberGuessingAp
4. Developed project repository: https://github.com/Uzair390-Del/NumberGuessingAppTflite.git